Technology Stack

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	3 Marks

Project Title

Personalized Travel Itinerary Planner

Frontend Technologies

Technology	Version	Purpose
React	19.3.x	UI component library for building interactive interfaces
TypeScript	7.0.x	Type-safe JavaScript for better developer experience
Vite	8.3.x	Lightning-fast build tool and dev server
TailwindCSS	4.3.x	Utility-first CSS framework for responsive styling
shadcn/ui	4.22.x	Accessible, customizable UI components (Button, Card, Input, Label, Dialog)
React Router DOM	7.18.x	Client-side routing and navigation
React Query (TanStack)	5.104.x	Server state management, caching, and data fetching
React Hook Form	7.90.x	Performant form handling with validation
Zod	4.6.x	Schema-based form validation
dnd-kit	6.3.x	Drag-and-drop functionality for activity reordering
Lucide React	1.60.x	Icon library
Supabase JS Client	2.118.x	Client-side Supabase Auth integration
Radix UI	1.8.x	Headless UI primitives for accessible components

Backend Technologies

Technology	Version	Purpose
Python	3.11+	Primary backend programming language
FastAPI	0.148.x	High-performance async web framework with auto-generated OpenAPI docs
Pydantic	2.14.x	Data validation and serialization using Python type hints
Pydantic Settings	2.16.x	Environment variable management with .env file support
Uvicorn	0.54.x	ASGI server for running the FastAPI application
Supabase Python	2.33.x	Server-side Supabase client for database operations
PyJWT	2.15.x	JWT token verification with JWKS support
HTTPX	0.28.x	Async HTTP client for calling LLM APIs
Pytest	9.1.x	Unit testing framework

AI / Generative AI Technologies

Technology	Purpose
Google Gemini API (gemini-2.5-flash)	Primary LLM for itinerary generation — fast, structured JSON output
Groq API (llama-3.3-70b-versatile)	Fallback LLM when Gemini is unavailable or rate-limited
Structured Prompting	Custom prompts enforce JSON schema output with real places and cost estimates

Database & Backend-as-a-Service

Technology	Purpose
Supabase	Backend-as-a-Service providing PostgreSQL, Auth, and realtime features
PostgreSQL	Relational database with full SQL support
Row-Level Security (RLS)	Database-level access control — users only see their own data
Supabase Auth	Email/password authentication with JWT tokens and JWKS key rotation
Supabase Project ID	vskaghlmvjqambnrshhy

Deployment & DevOps

Technology	Purpose
Vercel	Frontend hosting with automatic deployments from Git
Render	Backend hosting with environment variable management
Git / GitHub	Version control and source code management
GitHub Repository	https://github.com/Tanay-Ds/Personalized-Travel-Itenerary-Planner

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Architecture Summary

```
[Browser/Client]
       |
       |--- HTTPS --->  [Vercel - React Frontend]
       |                       |
       |                       |--- API calls (HTTPS) --->  [Render - FastAPI Backend]
       |                       |                                       |
       |                       |--- Auth --->  [Supabase Auth]       |--- DB queries ---> [Supabase PostgreSQL]
       |                                                               |
       |                                                               |--- LLM calls ---> [Gemini API / Groq API]
```